

4.0 DESIGN PROCEDURE IN PD.

DESIGN

In designing of a m/c Component, there is no rigid rule, problems may be approached in different ^{or} several ways, however, the general procedures involved in design are as follows:-

1. Identification of need — This constitutes a highly creative act, part of all, a complete statement of a problem, including the need, minor purpose for which a m/c is to be designed.

2. Definition of Problem — This is more specific which must include the specifications for the objects to be ^{designed} ~~designed~~. These specifications are the inputs & output quantities, characteristics & dimensions of the space to be occupied by the product.

3. Synthesis (mechanisms) — select the possible mechanisms or group of mechanisms which will give the desired motion.

4. Analysis & Optimization — This requires that a device model be constructed that will admit some form of mathematical analysis, this is often called a mathematical model.

5. Evaluation — This is the final proof of a successful design & usually involves the testing of a prototype in the laboratory. It is being tested here if the design really satisfies the needs, is it reliable? will it compete with similar product? is it economical to manufacture & to use? is it easy to maintain? etc.

6. Presentation — This is communicating the design to others, and is the final process.

presentation is a selling job

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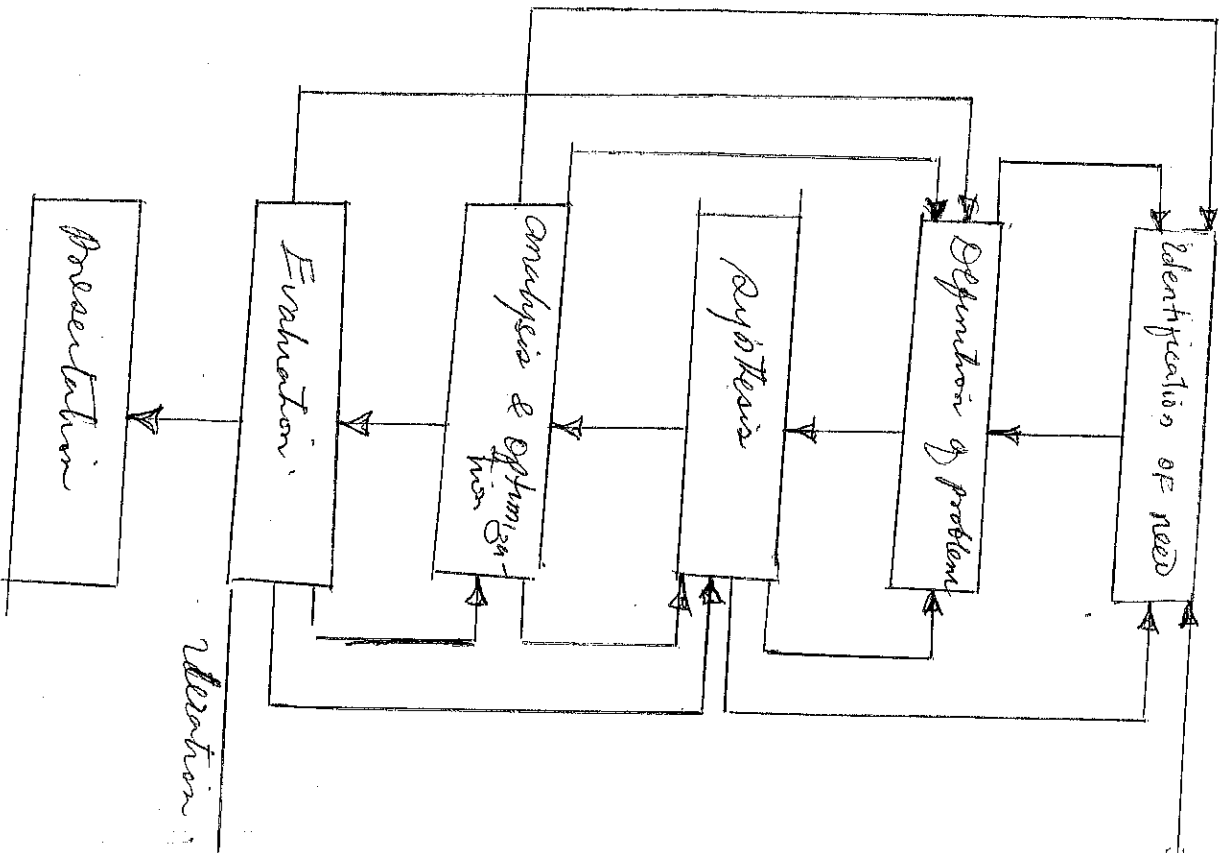


Fig. The phase in design, acknowledging the many feedbacks & iterations

3 BELT, ROPE & CHAIN DRIVES (1)

3.0 Belts, ropes, chains and other similar elastic or ~~plastic~~ M/c elements are used

in conveying options and in transmission

of power over comparatively long distances.

* ^{Advantage} These elements can be used ^{as a} replacement for gears, bearings, shafts & other rigid power transmission devices, they normally

simplify m/c design & and reduce cost.

* Because of their flexibility, ^{i.e.} ~~they~~ they are elastic and normally quite long, they absorb shock loads and damping out the effects of vibration but as a disadvantage, they don't have an infinite life, thus an inspection schedule has to be applied to guard against aging, wear & loss of elasticity which will lead to replacement.

3.1 Belts and ropes - There are two used to transmit power from one

Shaft to another by means a pulley(s) which rotate at the same speed or at different speeds. The power transmitted depends on the factors below:

1. Velocity of the belts
2. Tension under which the belt is placed on the pulleys.
3. The arc of contact b/w the belt and the smaller pulley.
4. The Condition under which the belt is used.

3.2. ~~Condition~~ Best Conditions under which belts should be used.

I Shaft should be properly in line to insure uniform tension across the belt section.

II Pulleys should not be too close so that the arc of contact on the smaller pulley may be as large as possible.

III Pollers should not be so far 2.

apart as to cause the belt to weigh heavily on the shaft which will increase friction on the bearings.

IV Long belts tends to swing from side to side, causing the belt to run out of the pollers, this develops crooked spots in belts

V. Tight side of belts should always be at the bottom so that weight even sag is present on the loose side will increase ~~friction~~ the arc of contact.

VI For good results on flat belts, ~~max~~ maximum distance of 10 metres ~~between~~ the pollers should not be exceeded, and minimum distance should not be less than 3.5 times the ~~diameter~~ diameter of the larger pulley.

3.3. Factors for Belt Selection

1. Speed of driving & driver shaft
2. Power to be transmitted.
3. Positive drive requirement.
4. Space available.
5. Speed reduction ratio
6. Centre distance b/w the shafts
7. Shafts layout.
8. Service conditions.

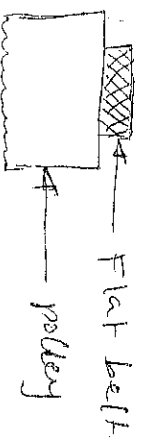
3.4. BELT DRIVE TYPES

- a. LIGHT DRIVES - These are for small power transmission at about ~~1000~~ 1000 ft/min speed of 1000 ft/min. eg Agric m/c & small m/c tools.
- b. MEDIUM DRIVES - For medium power with belt speed of over 1000 ft/min & up to 2200 ft/min. eg m/c tools.
- c. HEAVY DRIVES - For large power at belt speed of above 2200 ft/min. eg compressors & generators.

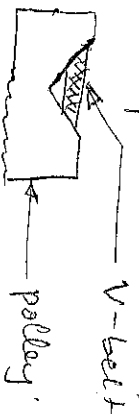
3.5 BELT TYPES - The most (3)

Common types of belts are

(a) Flat belts - Mostly used in factories and workshops, where a moderate amount of power is to be transferred from one pulley to another at the distance of not more than 8 metres apart.



b. V-Belt. Mostly used in factories & workshops where great amount of power to be transmitted, from one pulley to another, at a very near to another.



c. Circular belt - used at factories & workshops where a great power is to be transmitted from one pulley to another with a distance of more than 8 metres.



stretch at the same rate so that the load is equally divided between them.

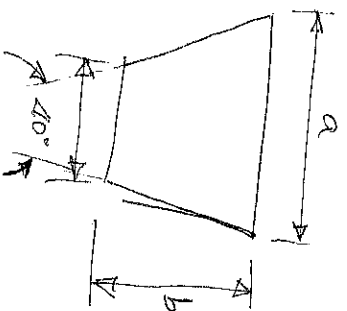
- If one of the belts set is broken, the entire set has to be changed at the same time, if one is just replaced, the new replaced will be more tighter & stretched as such will move with different velocity.

TYPES OF V-BELTS:

V-Belts are classified into five

types i.e. A, B, C, D, & E.

Type	width (a) mm	thickness (b) mm	Max. Pitch dia. mm	Power range in Kw.	weight per metre length in newton
A	12	8.5	75	0.2 - 7.5	1.06
B	16	11	135	0.7 - 18.5	1.89
C	22	13	230	11 - 75	3.43
D	30	19	325	37 - 186	5.96
E	38	25	540	75 and up	-



CHAINS

Chains are often used ~~for~~ to transmit motion + power from one shaft to another, when the center distance between their shaft is short. eg Bicycles, motor cycles, Agricultural m/cs, conveyor, rolling mills, road rollers etc. They can also be used for a center distance of up to 8meters, chains are used for velocities up to 25m/s & a power up to 110KW or more.

ADVANTAGES OF CHAINS OVER BELT/ROPE

1. No slip, hence, perfect velocity ratio obtained
2. Occupy less space in width than belts

(5)

3. It may be used for both long & short distances
4. higher transmission efficiency (upto 98%)
5. less losses on shafts
6. ability to transmit motion to several shafts by one chain.
7. Transmit more power than belts
8. Permits high speed ratios of 8 to 10 in a step
9. Can be operated under adverse temperature and atmospheric conditions.

DIS ADVANTAGES

1. Production cost is relatively high
2. Needs accurate mounting & careful maintenance.
3. High velocity fluctuation when runnig stretched.

Classification of Chains

Chains are classified into three groups based on their uses.

1. Hoisting & hauling (or crane) chain -

These chains are used for hoisting & hauling purposes & operate at the max. vel. of 0.25 m/s. They are either oval or

square in shape.

* The oval link chain has their joints welded & the sprockets used have receptacle to receive the links. They are used at low speed ~~well~~ ^{anchors} in hoist & in ~~anchors~~ for marine works.

* The square-link are used in hoist, crane dredges & has less manufacturing cost. i.e. the oval type.

2. Conveyor Chains. These type of chains are used for elevating & conveying

^{② Chain}
The material continuously at a speed upto 2 m/s. They are of 2 types. viz

* Detachable or hook joint type & closed joint type. Conveyor chains run at the speed of about 0.8 $\rightarrow$ 3 m/s.

3. POWER TRANSMISSION CHAINS

These type are used for transmission of power when the distance b/w the centre of shaft is short, they have provision for lubrication at ease, they are of 3 types.

- (a) Block or bush chain - This chain is in the early stages of development in power transmission. it produces noise when approaching & leaving the sprocket bites & rubbing between the teeth & links. Used as conveyor at a small speed.



5. Bush roller chain. — This type of chain

consist of outer plates or pin link plates, inner plates or roller link plates, pins, bushes & rollers. A pin passes through the bush hole is secured in the holes of the roller links the two sides of the chain. The rollers are free to rotate on the bush which protects

the sprocket-wheel teeth against wear

Advantage of bush roller chain

- They are extremely strong & simple to construct
- " give good service under severe conditions
- less noise due to impact of the rollers on the sprocket.
- used where there is little lubrication

C Silent chain — (inverted tooth chain)

These chain is designed eliminate the effects caused by stretching and to produce noiseless running.

(b) Brush roller chain -

Three types

Consist of outer plates or pin link plates, inner plates or roller link plates, pins, bush and rollers. A pin passes through the bush which is secured in the holes of the roller b/w the two sides of chain. The rollers are free to rotate on the bush with protects the sprocket wheel teeth against wear. The pins, bushes & rollers are made of alloy steel.